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The stress of travel

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The stress of travel

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International travel is an essential part of the life of elite athletes, both for competition and training. It is also becoming increasingly common among recreational sportspersons. Long-distance travel is associated with a group of transient negative effects, collectively referred to as 'travel fatigue', which result from anxiety about the journey, the change to an individual's daily routine, and dehydration due to time spent in the dry air of the aircraft cabin. Travel fatigue lasts for only a day or so, but for those who fly across several time zones, there are also the longer-lasting difficulties associated with 'jet lag'. The problems of jet lag can last for over a week if the flight crosses 10 time zones or more, and they can reduce performance and the motivation to train effectively. Knowledge of the properties of the body clock enables the cause of the difficulties to be understood (an unadjusted body clock), and forms the basis of using light in the new time zone to promote adjustment of the body clock. Sleep loss and its effects are important components of jet lag, and attempts to promote sleep by the use of melatonin and other hypnotics are also relevant. Sleep loss is also found in those who undertake challenges that involve long periods where the normal consolidated sleep of 8 h length is not possible. Advice on sleep regimens in such circumstances is given.

Keywords: jet lag, time-zone transition, travel fatigue.

The need for travel

Competitive sport is now recognized on a global scale and people at many levels have the opportunity of training and competing abroad. International competitions continue to proliferate (e.g. the qualifying rounds of the World Cup for soccer) and many sports involve international circuits, as in tennis, Formula 1 motor racing and golf; avid supporters need to travel widely to be keep in touch with their favourite team. At an intra-continental level, but still involving long journeys, teams and their supporters tend to travel widely. This applies to North American football, baseball and basketball teams, to Australian footballers and to many sports participants throughout Latin America, Europe and Asia. Furthermore, there are increases in recreational tourism, particularly skiers, underwater swimmers, surfers and golfers choosing to travel internationally to pursue their interests. Also, athletes might need to travel abroad for access to specialized training facilities.

International travel can be exhilarating and full of adventure, but it is also stressful and makes mental and physical demands upon all concerned – not only the athletes themselves, but also the support parties of coaches, trainers, physicians and physiotherapists. During and immediately after the day of travel, any long journey, nowadays generally a flight, will result in the traveller suffering from a group of symptoms which may collectively be called 'travel fatigue'. If the journey also entails crossing several time zones, then there is a longer-lasting group of symptoms popularly known as 'jet lag', the effects of sleep loss and loss of motivation to train being some of its main components. For those on journeys that take an extended period of time (e.g. yachtsmen sailing the Atlantic), the problem is not so much of the effects of time-zone transitions, since these are slow, but rather of accumulated sleep loss and physical endurance. In these cases, effective means of maintaining performance in the face of reduced sleep become a priority.

In all cases, individuals do not feel on peak form, and they perform sub-optimally, whether this entails competition, training or management of important tasks. For the holiday-maker or supporter, fatigue threatens to upset their enjoyment. The comparatively new science of chronobiology provides a framework for understanding many of the changes produced by travel, and this can become the basis for offering advice to minimize the negative effects.

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Since the 'body clock' is at the root of many of these problems, this review will deal first with the evidence for the existence of such a clock, and some of its properties that are relevant to the traveller. Then there will be an account of 'travel fatigue' – its causes and ways to ameliorate its effects. The main part of the review will deal with the problem of 'jet lag' and ways that have been attempted to alleviate it. The account will concentrate upon the methods that have a firm scientific foundation and have been formally tested. Even so, other methods will be mentioned. Finally, the issue of fragmented sleep, of particular relevance to long-distance rowers and those who sail yachts across oceans and round the World, will be covered briefly. The current review is timely in that a recent review of travel health information available at commercial travel websites concluded that the information was often inadequate (Horvath *et al.*, 2003).

Chronobiology

Chronobiology is the study of how living organisms respond to environmental time, the solar day (24 h), the lunar day (24.8 h), the monthly changes in the height of tides, and seasonal (yearly) changes. Daily rhythms are the most widespread and obvious, and the field has been surveyed by Reilly *et al.* (1997b).

In humans sleeping at night and active in the daytime, core temperature shows higher values in the daytime (peaking around 16:00–20:00 h) and lower values at night (minimum values around 03:00–06:00 h). Many other variables, including heart rate, mental performance, self-chosen work rate and several components of physical performance, show rhythms that are timed similarly. By contrast, for many hormones, including melatonin, and for the perceived exertion associated with mental or physical tasks, the observed rhythm is the inverse, with nocturnal values higher than those in the daytime (Figs. 1 and 2). The vast majority of physiological and biochemical variables show daily rhythms that fall into one of the above groups.

These rhythms do not result solely from responses to activity during the daytime and sleep and recuperation during the night. One of the clearest pieces of evidence for this comes from individuals studied during a 'constant routine' (Mills *et al.*, 1978). With this protocol, individuals are required to stay awake and sedentary for at least 24 h in an environment of constant temperature, humidity and lighting. They also need to engage in similar activities throughout this period, generally reading or listening to music, and to take identical meals at regularly spaced intervals. In these circumstances, any rhythmicity due to the environment and lifestyle has been removed; however, the rhythms of core temperature and other variables that have been studied persist, even though their amplitudes are decreased. The example of core temperature is shown in Fig. 3.

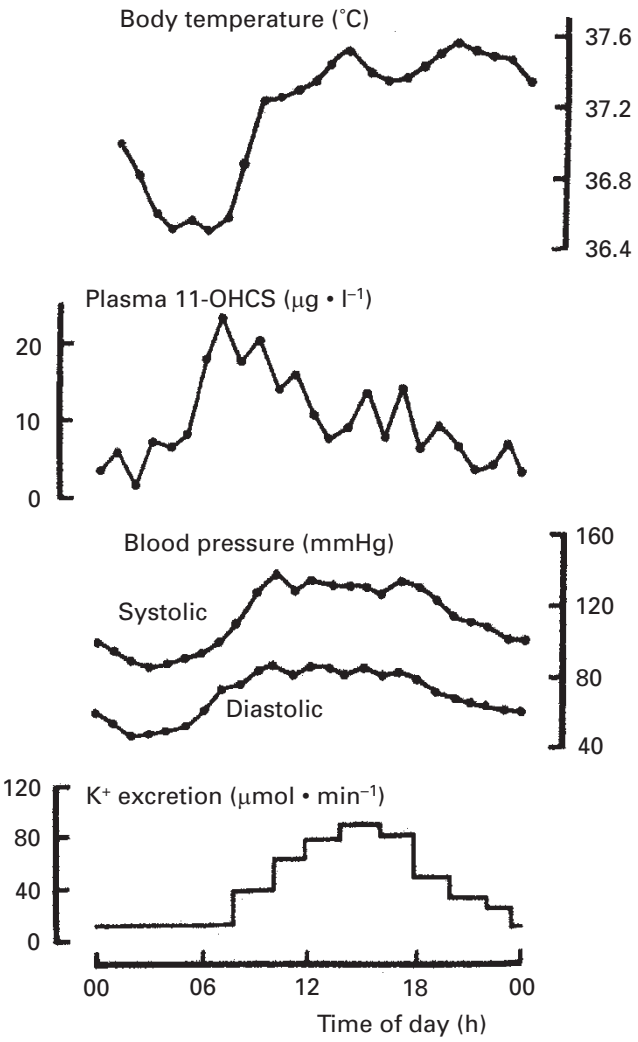


Fig. 1. Some circadian rhythms in an individual sleeping from 24:00 to 08:00 h (from Redfern *et al.*, 1991).

The rhythms that remain during constant routines arise from within the body and so have an endogenous component; this component is attributed to the so-called 'body clock'. When the temperature rhythm is measured in the presence of sleep and activity, effects of the environment and lifestyle are also present, as can be deduced from the finding that the amplitude of the rhythm is greater (Fig. 3). This component of the rhythm is termed 'exogenous'. In individuals living a conventional lifestyle, the two components are in phase. During the daytime, body temperature is raised by the body clock acting in synchrony with the alerting effects of the environment and physical and mental activities;

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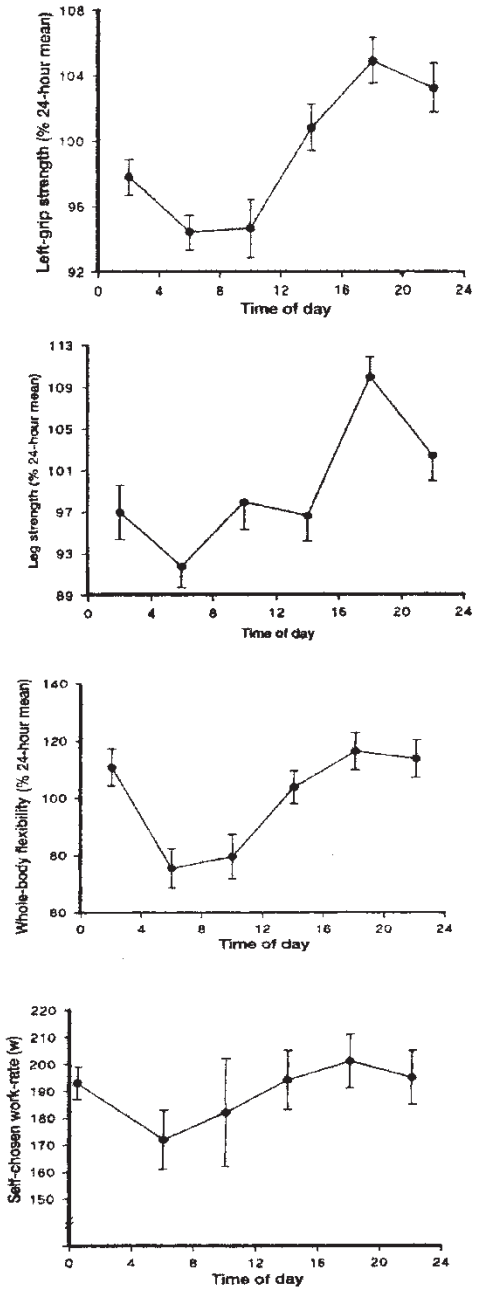


Fig. 2. Daily rhythms of (from top to bottom): grip strength, isometric knee extension, whole-body flexibility, self-selected work-rate (top three graphs from Reilly *et al.*, 1997b; bottom graph from Atkinson *et al.*, 1993).

during the night, the clock, environment and inactivity all act to lower body temperature. This interaction between the endogenous and exogenous components of a rhythm is found in all variables, though the nature of the exogenous component varies. For physical performance, the factors are the same as those for rectal temperature; for the melatonin rhythm, it is the alternation of light (which suppresses its secretion)

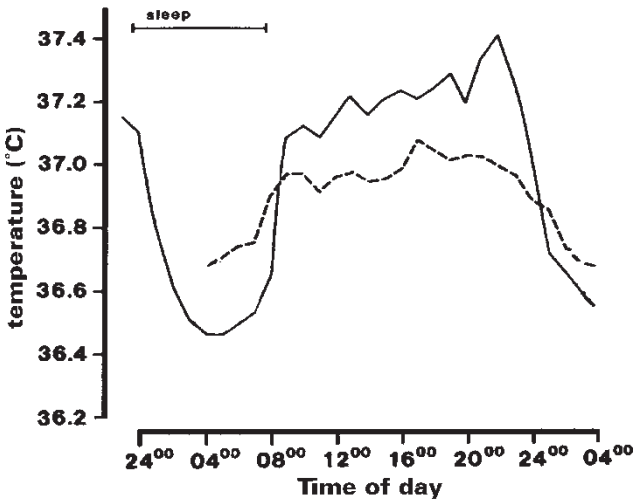


Fig. 3. Mean circadian changes in core (rectal) temperature measured hourly in eight individuals: living normally and sleeping from 24:00 to 08:00 h (solid line); and then being woken at 04:00 h and spending the next 24 h on a 'constant routine' (dashed line) (from Minors and Waterhouse, 1981a).

and dark; for gut activity, it is the alternation of feeding and fasting; for the secretion of growth hormone, it is the first bout of 'deep' (slow-wave, Stages 3 and 4) sleep; and for mental performance there is an effect of fatigue, by which performance falls with an increasing amount of time spent awake.

The site of the body clock and its circadian nature

Paired groups of cells are found in the human in the base of the hypothalamus, a region of the brain closely associated with temperature regulation, the release of hormones, appetite for food and sleep. These cells are known as the suprachiasmatic nuclei (SCN; Moore, 1995). Cells from the SCN of mammals show a daily rhythmicity in nerve activity when they are cultured *in vitro* in constant conditions; no other cells from the brain show autonomous activity in such circumstances. The conclusion from this piece of evidence (there are many others) is that the SCN is the site of the body clock.

Recent work upon plants and animals, including humans, has indicated that there is a large degree of evolutionary conservation with regard to clock mechanisms, their molecular biochemistry and their genetics. In brief, a group of 'clock genes' in the nucleus is converted in the cytoplasm of cells into a group of 'clock proteins' (see Clayton *et al.*, 2001; Miyazaki *et al.*, 2001). These proteins are responsible for many of the rhythmic processes (neural activity, transmitter production, and so on) associated with the cells of the SCN. After modification in the cytoplasm, some of these

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proteins move back into the nucleus where they inhibit the activity of the clock genes. This results in a decline in production of the clock proteins, which, in turn, means that there is a loss of inhibition of the clock genes. Therefore, the clock genes become active once again and so a new cycle begins. The whole cycle takes about 24 h to be accomplished.

When humans are studied in an environment free from time cues – in an underground cave, for example – the daily rhythms of sleep and waking, body temperature, hormone release, and so on continue. This fact supports the view that they have endogenous components, but the period of such rhythms is closer to 25 h than 24 h (Wever, 1979). As a result, an individual becomes progressively more delayed with respect to the outside environment. For this reason, the 'free-running' rhythms measured in such conditions are called circadian (from the Latin for 'about a day').

Adjustment of the body clock

To provide useful information about solar time, this circadian clock needs to be synchronized to a solar (24-h) day. Synchronization is achieved by *zeitgebers* (German for 'time-givers') – rhythms resulting, directly or indirectly, from the environment. When the plant and animal kingdoms as a whole are considered, the rhythms of the light–dark cycle, of feeding–fasting, of activity–inactivity and of social influences, acting singly or in various combinations, all act as *zeitgebers* for certain species. For humans, the most important *zeitgebers* are the light–dark cycle and the regular release into the bloodstream of melatonin by the pineal gland (Arendt, 1992) during the hours of darkness.

Melatonin secretion and the rhythm of core temperature are closely linked temporally. Melatonin has several effects upon the body, including a cutaneous vasodilatation that increases heat loss (Atkinson *et al.*, 2003). In the evening, therefore, core temperature falls due to the combined effects of the rise in melatonin secretion, the body clock re-setting thermoregulatory mechanisms, and individuals relaxing in a quiet environment. In the morning, changing thermoregulatory mechanisms due to the body clock, the termination of melatonin secretion, and beginning activity in a more dynamic environment, all act to raise core temperature.

Bright light acts as a *zeitgeber* because its effect upon the body clock depends on the time at which an individual is exposed to it (Czeisler *et al.*, 1989; Minors *et al.*, 1991). Pulses of bright light (as might be found outdoors) that are in the 6-h 'window' immediately after the trough of the body temperature rhythm (the trough normally being 03:00–06:00 h, see Figs. 1 and 3) produce a phase advance; pulses in the 6-h window before the temperature minimum, a phase delay; and those away from the temperature trough by more than about 6 h have little effect. Weaker light pulses (similar in intensity to domestic lighting) produce smaller phase shifts (Boivin *et al.*, 1996; Waterhouse *et al.*, 1998; Boivin and James, 2002); this last observation is relevant to the many humans who work and live indoors and use curtains and artificial lighting, and so have little exposure to the natural alternation of daylight and darkness. In practice, therefore, exposure to light on waking in the morning, and even to light passing through our eyelids when asleep at this time, will cause a small advance of the body clock. This daily advance offsets the fact that the body clock shows a circadian period slightly in excess of 24 h, and results in the circadian clock becoming synchronized to the solar day.

When light acts as a *zeitgeber*, information passes to the SCN from the eyes via a direct pathway, the retinohypothalamic tract. It seems that the retinae contain a sub-group of receptors whose visual pigment is based on vitamin B_2 (rather than the more conventional vitamin A-based opsins), and that it is this sub-group that is responsible for the light signals that go to the SCN (Miyamoto and Sancar, 1998).

Ingestion of melatonin can also advance or delay the body clock, according to the time of ingestion (Lewy *et al.*, 1998). The shifts produced at a particular time are in the opposite direction to those produced by light exposure at that time; thus, melatonin ingestion in the afternoon and evening tends to advance the body clock and, in the morning, to delay it. Because of these effects of melatonin upon the body clock, and the fact that it is secreted into the bloodstream each night, melatonin is sometimes called an 'internal *zeitgeber*'. Receptors for melatonin have been found in the SCN, and are believed to be the means by which it acts upon the body clock. Since bright light inhibits melatonin secretion, the phase-shifting effects of light and melatonin reinforce each other. Bright light in the early morning, just after the temperature minimum, advances the phase of the body clock – not only directly via the retinohypothalamic pathway but also indirectly, since it suppresses melatonin secretion and so prevents the phase-delaying effect that melatonin would have exerted at this time (Lewy and Sack, 1996).

Roles of the body clock

The body clock transmits its rhythmic output to the hypothalamus and many other regions of the brain. Since these areas control temperature regulation, hormone release, feeding and sleep, it becomes clear that the physiology, biochemistry and behaviour of the whole body will become rhythmic, with a phase determined by the SCN.

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This widespread rhythmicity, extending throughout the body, serves two functions. First, it shifts the body towards an 'active' mode in the daytime – encompassing physical and mental activities, and respiratory, metabolic and cardiovascular changes – and towards recovery and restitution in the night – involving sleep and changes to all the variables associated with the active mode. Since such changes cannot be achieved immediately, but require to be initiated before the actual events of waking up or falling asleep occur, it is the second role of the body clock to enable such preparations to be made. During the evening, the body clock, normally aided by an individual's habits and environment, alters their behaviour, physiology and biochemistry to prepare them for sleep; towards the end of sleep, changes in physiological and biochemical variables, often in concert with increased environmental light and noise, prepare them for waking up.

All these rhythms in physiology and biochemistry are of direct relevance to everybody, including athletes in competition and during their normal training programmes (Reilly *et al.*, 1997b). The abilities to fall asleep and to stay asleep are controlled by the body clock, being easier when core temperature is falling (in the evening) or low (in the night). When awake, individuals' adrenaline secretion, muscle strength, short-term maximal power output and self-chosen work rate are all greatest at about 18:00 h (when core temperature is around its peak). At about this time also, many aspects of mental and physical performance are at their best (Fig. 2), and the perceived exertion for a given amount of physical and mental activity is felt to be least. Most personal best performances in sport are set at this time, although this phenomenon is doubtless influenced in part by event-scheduling.

In summary, our current understanding of chronobiology indicates that we are rhythmic animals attuned to a rhythmic world, and that regular activities and a regular environment play important parts in this integration. During travel, however, particularly over long distances, these regular habits are thwarted by changes to our normal schedules, and this disruption contributes to 'travel fatigue'.

Travel fatigue: the effects of long-distance journeys with little change in the time zone

The symptoms of travel fatigue are shown in Table 1, and can be summarized by the term 'travel weariness'. Travel fatigue is caused by several factors in addition to the temporary change in habits that a long-distance

Table 1. Checklist for travel fatigue (based on Waterhouse *et al.*, 2002b)

Symptoms
Fatigue
Disorientation
Headaches
'Travel weariness'
Causes
Disruption of normal routine
Hassles associated with travel (checking in / baggage claim / customs clearance)
Dehydration due to dry cabin air
Advice
Before the journey:
Plan the journey well in advance
Try to arrange for any stopover to be comfortable
Make sure about documentation, inoculations, visas, etc.
Make arrangements at your destination
On the plane:
Take some roughage (e.g. apples) to eat
Drink plenty of water or fruit juice (rather than tea/coffee/alcohol)
On reaching the destination:
Relax with a non-alcoholic drink
Take a shower
Take a *brief* nap, if required

journey requires. These additional factors include preparations for the flight, the flight itself and adjusting to the new environment. Many of these problems can be minimized with sufficient forward planning. Provided appropriate remedial action is taken (Reilly *et al.*, 1997b), the symptoms disappear within 48 h.

Advice before the flight

Preparations for the journey include such incidentals as ensuring that the passport is valid, that any visa requirements have been met, that health advice has been sought and acted upon, that there are sufficient funds for the trip, and that any medication that might be required has been packed. In the case of professional teams, responsibility for such details is assumed by the team's management. Taking suitable clothing is most important; sufficient and appropriate clothing should be taken for the journey itself, for training and for the competition itself, and for relaxation.

Preparations also include arranging the flight or flights, and it is here that often choices can be made about whether or not to fly by day or by night, and the length of any stopovers *en route* (Tsai *et al.*, 1988). It should be ensured that there is sufficient time if connecting flights are involved; hanging around might not be a desirable way to spend time, but it is far less stressful than having to rush around and hope that connections will be held if there are delays. Arranging travel early might also enable seats with more leg-room (e.g. those near the emergency exits) to be reserved.

On the day of the journey itself, the traveller is advised to start early and try to arrive at the airport with plenty of time to spare. This will ensure that problems do not arise if delays do occur or if more time is required for extra security arrangements.

Being properly prepared helps to allay anxiety before travel (Bor, 2003). Indeed, our observations on travellers to Australia from the UK confirm the value of arriving at the airport refreshed (Edwards *et al.*, 2000).

Advice during the flight

The flight cabin, particularly in economy class, is a cramped area and provides only a very limited opportunity for movement. Being immobile for extended periods of time increases the risk of developing cramp and even, with longer journeys, deep-vein thrombosis in susceptible individuals: this condition has been referred to as 'economy class syndrome' (Mercer and Brown, 1998; Kraaijenhagen *et al.*, 2000; Brown *et al.*, 2001). Passengers are advised to walk around the cabin and to do some stretching exercises where and when possible. Isometric exercises in the seat can liven up a sluggish circulation, and it has been suggested that wearing support stockings might be beneficial (Belcaro *et al.*, 2001).

The cabin air is very dry; this low humidity leads to a general dehydration, with lips and eyes becoming sore and the mouth and throat becoming dry. Sore lips can be avoided by the use of lip salve. For people who usually wear contact lenses, it is suggested they wear spectacles instead. Dehydration can be opposed by increasing fluid intake, but alcoholic drinks, tea and coffee are excluded from this advice. These drinks tend to cause further dehydration due to their diuretic effects; spring water, soft drinks and fruit juice are all more suitable.

There is always the problem of deciding how to pass the time on a long flight. Sleeping or napping is advised only when it is night at the destination, or if the individual has been chronically deprived of sleep during transit. If too much sleep is taken during the flight when it is daytime at the destination – if only because of boredom – the traveller might find it difficult to sleep at night after arrival at the destination. When the aim is to avoid sleep, the traveller should find ways of maintaining arousal such as talking to fellow passengers, reading, playing cards or making use of the in-flight entertainment that is provided.

Advice after the flight

After arrival, baggage has to be collected, customs must be cleared and the final destination must be reached, all of which take time and can cause aggravation, particularly if any problems arise. There might also be the added problem to contend with of unfamiliarity with the language. When the destination is finally reached, the tendency is to sleep because of fatigue and 'winding down'. This practice is reasonable when it is night-time, but this is not often the case. Having arrived in the daytime, a recuperative nap of about one hour, followed by an envigorating shower and relaxation, are recommended instead (de Looy *et al.*, 1988).

By the next day, most of the effects of travel fatigue will have worn off, and the traveller can continue with preparations for competition and/or enjoy the new environment. The problems and advice to travellers are summarized in Table 1.

Jet lag

For long-distance flights across several time zones, problems in addition to travel fatigue arise. The problems stem from the properties of the body clock, and an understanding of these also becomes relevant when there is a need to minimize the difficulties.

As already described, the body clock promotes activity in the daytime and uninterrupted sleep at

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night. To achieve this regularly – in people living normally, for example – requires the body clock to be stable; it must not respond to (that is, be adjusted by) transient changes in the environment or lifestyle of the individual. A clock that rapidly adjusted to these changes would compromise the timing of the circadian rhythms of an animal during an eclipse of the sun, of an individual who woke transiently in the night, or of one who took a nap in the daytime. The observation that the body clock is, in fact, slow to adjust to changes in lifestyle makes sound ecological sense, therefore (Waterhouse, 2001). Recently, it has been shown that the daily rhythms associated with the production of some of the 'clock proteins' in the SCN of rats adjust only slowly to a simulated time-zone transition (Nagano *et al.*, 2003).

Nevertheless, such stability becomes the bane of those who undergo time-zone transitions since, before adjustment of the body clock to the changed local time has taken place, the normal synchrony between the endogenous and exogenous components of the circadian rhythms (see Fig. 3) is lost. Some of the problems faced, particularly marked when sleep and waking activity are considered, are shown in Table 2.

Symptoms of jet lag

The negative symptoms and other effects associated with this lack of synchrony between the body clock and the outside world are collectively known as 'jet lag' (Reilly *et al.*, 1997b; Waterhouse *et al.*, 1997). The symptoms are not due to differences in climate or culture between the destination and the individual's home country. For example, they are likely to be marked for Europeans travelling to Australia, New Zealand or the west coast of America, but slight for travellers between Europe and Africa.

The symptoms of jet lag most commonly reported are listed in Table 3. The wide spectrum of negative effects implies that most individuals might be affected in a way that is particularly important to them at a personal level. Athletes, for example, might be discouraged by poorer performances in the new time zone and less motivation to train hard.

Problems associated with subjective assessments of 'jet lag'

Much current research on 'jet lag' leaves the interpretation of the exact meaning of this term up to the participants themselves. Moreover, subjective assessments can be made of one or more symptoms of jet lag – of 'fatigue', for example – but the link that exists between the assessment of 'jet lag' and of one or more of its symptoms is unclear. As a result, comparisons between studies can be difficult. A more systematic approach to the problem was adopted by Spitzer *et al.* (1999), who showed that there was a high internal consistency between assessments of the following items: 'fatigue', 'difficulty concentrating', 'clumsiness', 'decreased alertness in the daytime', 'difficulty with memory', 'general weakness', 'dizziness', 'lethargy' and 'daytime sleepiness'.

Recently, we have considered some of the problems that exist during assessments of jet lag (Waterhouse *et al.*, 2000, 2003a). If the symptoms shown in Table 3 were equally valid assessments of jet lag, then it is predicted that their time-courses of recovery after a time-zone transition would be similar. This was not found to be the case in a study of a group of athletes who flew from the UK to Australia, 10 time zones to the east (Waterhouse *et al.*, 2000). They reported the severity of jet lag at five times spread regularly throughout the waking day for the first 6 days after arrival in Australia. In addition, they recorded regularly their fatigue, enjoyment of meals, bowel habits, mental performance and subjective response to aspects of sleep. Mean daily assessments of the changes from

Table 3. The main symptoms of jet lag (based on Waterhouse *et al.*, 1997)

- Feeling tired in the new local daytime, and yet unable to sleep at night
- Feeling less able to concentrate or to motivate oneself
- Decreased mental and physical performance
- Increased incidence of headaches and irritability
- Loss of appetite and general bowel irregularities

Table 2. Some differences from living in the home time zone that will be experienced in the first days after flying across eight time zones to the east or west (based on Reilly *et al.*, 1997b)

Local time	Home time zone	After 8 time zones east	After 8 time zones west
08:00–10:00 h	Waking up	Ready to go to sleep	Going well
14:00–16:00 h	Going well	Beginning to wake up	Ready to go to sleep
20:00–22:00 h	Preparing to wind down	Going well	Want to be asleep
02:00–04:00 h	Sleeping well	Going well – not sleepy	Woken up and ready to go

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normal of all these variables were most marked on the day of arrival, improved on subsequent days, and had recovered, or almost so, by day 6. However, the time-courses of recovery differed between the various symptoms. Whereas increased fatigue, decreased ability to concentrate and problems with some aspects of sleep (waking too early and the increased number of waking episodes during sleep) all recovered at rates similar to that of the decline of jet lag, other variables (e.g. decreased hunger and increased irritability) recovered with shorter time-courses.

A further analysis (Waterhouse *et al.*, 2003a) considered the relationships between assessments of jet lag and its symptoms, and whether these changed at different times of the day. For example: on waking, jet lag might reflect mainly the perceived inadequacy of the previous night's sleep; during much of the daytime, it might be more likely to reflect increased fatigue and decreased motivation; and, when retiring at night, jet lag might reflect an inappropriate feeling of wakefulness at this time. It was found that some measurements of sleep disruption correlated highly with estimates of jet lag made either just before retiring at night or just after rising in the morning, but not significantly with estimates of jet lag made in the middle of the daytime. Also, there was a strong positive correlation between estimates of jet lag made at any time of the day and fatigue measured at the same time, but not with fatigue measured at any other times of the day. The associations between mental performance or the enjoyment of meals and jet lag were statistically weaker and applied equally well to estimates of jet lag made at wider ranges of times during the day. That is, the links between jet lag and its symptoms varied in strength and in temporal specificity.

Collectively, these findings are difficult to reconcile with the view that all the symptoms of jet lag are equally valid descriptions of jet lag, both when different days after the flight and when different times during the day are considered. Such difficulties might have important implications when the efficacy of methods to alleviate jet lag are investigated, since there is often the implication that alleviating 'jet lag' is reducing its other symptoms, and that the effect applies at all times of the day.

Differences between individuals in their experience of jet lag

In general, the symptoms of jet lag are worse as the number of time zones crossed increases, and on travelling eastwards rather than westwards (Minors and Waterhouse, 1981a; Reilly *et al.*, 1997b). Individuals also experience the symptoms of jet lag to differing extents, and the severity of the symptoms may be related to menstrual cycle phase. Disturbances of this cycle, including secondary amennorhoea, are not uncommon in air stewardesses (Suvanto *et al.*, 1993). The disturbances might arise from an interaction between melatonin secretion (which is affected by the light–dark cycle) and oestrogens released during the menstrual cycle. Changes to melatonin secretion in female flight attendants have recently been reported by Grajewski *et al.* (2003).

It has been suggested (see summary in Minors and Waterhouse, 1981a) that several characteristics of individuals might alter their rate of adjustment to a time-zone transition. The characteristics include age, chronotype, sleeping habits, languor and fitness. The predictions are:

i *Age*: younger individuals are more flexible and will have less difficulty.
ii *Chronotype*: 'larks', or morning types, will adjust to an eastward shift more readily than 'owls', or evening types, since their body clock might have a slightly shorter inherent period. The opposite should hold for time-zone transitions to the west.
iii *Flexibility of sleeping habits*: people who are better able to adjust their times of sleeping, and who are influenced less by the conditions in which they sleep (the hardness of the mattress, for example), should be at an advantage.
iv *Languor*: individuals who score highly on this measure have an inability to disregard feelings of fatigue, and experience difficulties with continuing their activities effectively if they feel fatigued. They will be disadvantaged.
v *Fitness*: people who are physically fitter should experience less difficulty with adjustment.

The experimental support for these predictions is weak, however. In a further analysis (Waterhouse *et al.*, 2002a) of the results for the individuals who flew from the UK to Australia (Waterhouse *et al.*, 2000), it was investigated if the above characteristics of individuals acted as predictors for the amount of jet lag and its symptoms experienced after the flight. To these possible predictors were added gender, experience of long-haul travel, time of arrival in Australia (the morning or late afternoon by local Australia time) and direction of phase adjustment (advance or delay).

The only reliable predictors were previous experience of long-haul travel, age and time of arrival in Australia. Those for whom such a long-haul flight was a new experience tended to retire later and suffered more jet lag. The finding that age was associated with less jet lag was the opposite of that predicted (above), but accords with the view from laboratory simulations that older individuals are better able to 'pace themselves' when dealing with the effects of sleep loss (Moline *et al.*,

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1992). Those people who arrived in Australia in the afternoon (and had left the UK in the morning) suffered less jet lag than those who arrived early in the morning (and had left the UK in the evening). This decrease in the amount of jet lag suffered was in spite of their having had significantly less sleep during the first flight, which had been in the daytime. The advantage for these individuals appears to have been that they were able to attempt their first full sleep in a comfortable bed during the Australian night about 30 h after having risen from their last sleep at night in a bed in the UK. In contrast, those who had arrived in the morning by Australian time, even though they had taken a nap on arrival and had slept more during the flights, did not have the opportunity for a full sleep in a bed at night in Australia until about 50 h after having woken from their last sleep at night in a bed in the UK.

Waterhouse *et al.* (2002a) also found that jet lag was worse on days 5 and 6 after arrival in those who were adjusting their body clock to the new time zone by a phase advance. For all individuals immediately after arrival in Australia, their temperature minima and worst performances were at about 15:00 h by local time (05:00 h UK time). Adjusting by phase delay meant that the time of temperature minimum and poorest performance became later, soon coinciding with the evening and the time for relaxation. By contrast, for those who were adjusting their body clock by a phase advance, the times of minimum temperature and worst performance advanced through the early afternoon and then the morning. This would be noticeable, since it would coincide with the time when concentration and physical effort were required, and so would account for the greater feelings of jet lag on the fifth and sixth days after arrival.

In summary, these studies (Waterhouse *et al.*, 2000, 2002a, 2003a) indicate that the choices of itinerary and lifestyle are more important for reducing jet lag than are personality characteristics such as chronotype or flexibility of habits. Moreover, if disruptions during the working day are to be minimized, then there is evidence to support attempts to adjust to the new time zone by promoting a delay rather than an advance of the body clock. However, it must be pointed out that there are very few field studies in which objective measures of the phase of the body clock have been recorded. The reasons for this are that the problems posed by measuring accurately markers of the body clock – rectal temperature during constant routines or melatonin secretion during dim light, for example – are difficult to solve. However, it has been shown recently (Carvalho Bos *et al.*, 2003) that suitable treatment of the actimetry record can give useful information in similar conditions, and this approach might prove to be productive.

Effects of sleep loss and jet lag upon mood and mental performance

One of the main difficulties associated with jet lag is sleep loss (Gander *et al.*, 1993). After flights to the east, it is getting to sleep at the new (apparently earlier) bedtime that is difficult, whereas after flights to the west it tends to be premature awakening (Table 2). Sleep loss leads to an increased risk of decrement in mental performance, and it has been suggested that jet lag might be associated with errors or accidents in aircrew (Akerstedt, 1995; Dinges, 1995). The results of Samel *et al.* (1995) support this view; particularly with flights to the east at night, pilot fatigue increased during the flight to values that were likely to be associated with decrements in mental performance. A recent study stressed that fatigue was present in both long-haul and short-haul pilots, which was attributed by the pilots to a combination of jet lag and night flights in the former group (Bourgeois-Bougrine *et al.*, 2003). The negative effects of sleep loss upon mood and mental performance have been found in far wider contexts – for example, in field studies of military personnel and civilians, and in laboratory studies with volunteers (Dinges, 1995; Rosa, 1995). Such deterioration can adversely affect the interactions between individuals and their individual motivation; it is likely to be important for games players and athletes in training, therefore.

Athletic performance following time-zone transitions

By contrast, muscle strength and all-out physical activity have generally been found to be resistant to any deterioration after sleep loss, although self-chosen work rate and rise in perceived exertion may become altered in sleep-deprived individuals (Reilly *et al.*, 1997b). The longer the performance, the more likely is it to be affected by sleep loss. For example, execution of maximal and sub-maximal lifts was found to deteriorate during the course of a typical weight-training session in individuals partially deprived of sleep (Reilly and Piercy, 1994). A deterioration in performance of indoor football players in 'marathon' exhibitions, where sleep is lost, has been reported (Reilly, 1994). In these, activity is continuous and has to be sustained, and it seems that, as with mental performance, deterioration is considerably greater when there is only limited opportunity for short rests to be interposed.

For competitors in international competition, immediately after time-zone transitions there generally is a decline in performance of activities involving complex mental activity, together with a feeling of lethargy and a general loss of motivation (Reilly *et al.*, 1997b). These problems are due not only to increased fatigue and

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negative effects on mood and physical and mental performance produced by sleep loss, but also to a lack of adjustment of the body clock, as a result of which the athlete is awake during what, by 'body time', is the night. Maximum performance might be lessened if, as is normally the case, it does not coincide with the individual's circadian peak, and serious training might also be jeopardized due to its inappropriate timing. As Table 2 indicates, in the days immediately after a time-zone transition, this coincidence between the time of training and the circadian peak of an unadjusted body clock will not exist.

An increased time taken for both a sprint and a middle-distance run after an eastward flight crossing six time zones has been shown (Wright *et al.*, 1983). Also, British rugby league players who travelled to Australia showed a disturbance in their circadian rhythm in grip strength, which normally peaks in the evening (Reilly and Mellor, 1988). Only by the fourth day were strength scores in the evening higher than the morning scores, and it took 2–3 more days for the normal circadian rhythm to restore itself. Similarly, leg strength showed a circadian rhythm disturbance in a group of athletes on the first day after a westward transition across five time zones (Fig. 4); the time of peak strength did not shift to its normal value of about 17:00 h until about 5 days after the flight.

Training programmes employed during the period immediately after a time-zone transition should acknowledge these negative effects, and be reduced accordingly (Lemmer *et al.*, 2002). Moreover, reduced levels of achievement should be anticipated by the coach as well as the athlete or player during this time of adjustment.

Coping mechanisms for dealing with time-zone transitions

The aim of all concerned is to minimize the problems associated with time-zone transitions. The advice depends upon whether the transition is made slowly or rapidly and, in the latter case, upon the duration of the stay in the new time zone and the direction of the flight. Many Olympic and international teams receive advice about coping with travel fatigue and jet lag but guidance is not always based on scientific rationale. Therefore, the evidence for various travel strategies is considered in the sections that follow.

Slow transitions across time zones

When an ocean liner or car is used to travel long distances, any time-zone transitions are accomplished slowly. At the equator, one time zone is approximately 1000 miles (1600 km), which means that the traveller will cross less than one time zone per day. This should produce very little inconvenience or jet lag, the body clock being required to be advanced (eastward journey) or delayed (westward journey) by less than an hour each day. The effect would be comparable to that produced when local time is adjusted by 1 h in the switches from winter time to Daylight Saving Time, for example (see Valdez *et al.*, 2003).

For those who sail or row across oceans, there is the difficulty of getting enough sleep in an environment in which continuous alertness and effort are required. Laboratory studies indicate that sleep loss of as little as 2 h can lead to a deterioration in some aspects of mental performance (Graeber, 1989; Dinges, 1992; Naitoh *et al.*, 1993). The decrements increase with the total amount of sleep lost, and tend to be worse for repetitive tasks and when the core temperature is near its trough. 'Micro-sleeps' might occur – individuals appearing to be awake but unaware of what is happening around them (Folkard, 1990). Physical activities are less affected by sleep loss, but general mood and motivation to work will be affected adversely, as will coordinated actions within a group of individuals. Sleeps of as little as 1–2 h per 24 h are able to

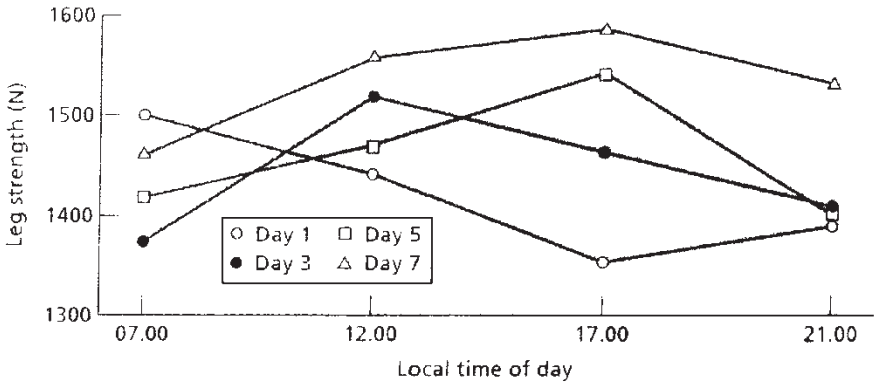


Fig. 4. Change in local timing of leg strength (Newtons) on days 1, 3, 5 and 7 after a westward flight across five time zones (based on Reilly *et al.*, 1997b).

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ameliorate some of the effects, however (Naitoh, 1992; Rosekind *et al.*, 1995).

There has been a tendency for those facing the prospect of prolonged effort and potential sleep loss to change from the normal pattern of 8 h sleep and 16 h of waking each 24 h. Instead, the practices of living shorter 'days' and 'power napping' have developed – for example, dividing the 24 h into two 'days' of 4 h sleep and 8 h waking, or four 'days' of 2 h sleep and 4 h waking (Stampi, 1992). These retain the normal balance between times awake and asleep, and have the advantage that the periods of activity (which causes fatigue) and when the individual is asleep (not contributing to the overall effort of sailing or rowing, or oblivious of the environment) are both decreased.

Such schemes do not result in any obvious adverse side-effects when implemented for several days. The evidence indicates that sleep in its various stages continues to be taken, and that fatigue builds up less during the shorter periods of activity (Stampi, 1992). There does appear, however, to be a limit to how short the 'day' can be, this restriction being determined by the abilities to take recuperative, deep (slow wave, Stages 3 and 4) sleep and to overcome the effects of 'sleep inertia' immediately after waking (Dinges, 1992; Naitoh, 1992; Naitoh *et al.*, 1993). For these reasons, sleeps much shorter than 2 h might be of only limited value in the long term.

It is important to realize that laboratory-based tests of the effects of sleep loss and of different sleep–wake regimens have been performed over periods of days or a few weeks rather than over months, and that the possibility that the demands of rowing and navigation act as antidotes to sleep loss has not been tested. Finally, laboratory-based studies cannot incorporate the elements of danger or excitement that these tests of endurance possess.

Brief stays (3 days or less) after a flight to the new time zone

When the stay in the new time zone is for a few days only, there is insufficient time for the body clock to adjust. In these cases, it is common for many travellers, including aircrew, to continue to live as much as possible on home time (Lowden and Akerstedt, 1998). For the businessperson, this means that, after a flight to the east, important activities should be undertaken in the afternoon or evening by local time in the new environment (coincident with daytime on home time), and the morning (coincident with night on home time) should be left free for relaxation or sleep. For similar reasons, if the flight is to the west, then important activities should be arranged for the morning, with the afternoon and evening being reserved for relaxation and sleep (Waterhouse *et al.*, 2003b).

Such arrangements are unlikely to be satisfactory for athletes, who will probably suffer from some degree of sleep loss. Also, there will not be enough time to settle into a new routine, and no familiarity with the environment in which competition is to take place will have been obtained. Longer stays before serious training and an important event are strongly advised, therefore.

Longer stays (5 days or less) after a flight to the new time zone

With longer stays in the new time zone, adjustment of the body clock to local time should be the aim. The following advice is the best currently available for reducing to a minimum disruptions to sleeping and decrements in performance in the new time zone. The advice applies particularly to the first day or so in the new time zone, some of it applies as soon as the home time zone has been left behind, and some applies even when plans for the flights are being made.

Before travel

A search to find the most convenient travel schedules is encouraged. Jet lag is less if the total time that elapses between the last proper sleep in the country being left and the first proper sleep in the destination country is as short as possible (see above). Also, itineraries involving a stopover *en route* of a day or so are worth considering because simulation studies have indicated that dividing the total time-zone transition into more than one sector (for example, a 10-h shift into shifts of 7 h and 3 h) results in a decrease in the severity of jet lag (Tsai *et al.*, 1988). In practice, however, the advantages of such a stopover should be weighed up against increased costs and increasing any stresses due to baggage transportation, passport controls and searching for accommodation. Attempts have been made to pre-adjust partially to the time zone to be visited, by going to bed 1 h earlier each night before a flight to the east, or 1 h later each night before one to the west. These have been shown to produce partial adjustment of the body clock in the appropriate direction, particularly if there is also appropriate timing of light exposure (Burgess *et al.*, 2003). However, the usefulness of this approach might be limited by the inconvenience and difficulty associated with altering the hours of sleep in the days before the flight.

During travel

Advice regarding the dry cabin atmosphere, the cramped conditions and whether or not to sleep applies

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equally as when combating travel fatigue. A further good piece of advice is to realize that, as soon as the plane has been boarded, 'home time' has been left behind and should be erased from the traveller's mind; instead, it is the time at the destination or stopover point that now determines when things should be done. Changing one's watch immediately on boarding the plane helps to achieve this attitude of mind.

Some travellers have used minor tranquillizers, such as temazepam, Zolpiden and other members of the diazepine group, to help them be refreshed for immediate activities on arrival (Atkinson *et al.*, 1997). The evidence for sleeping pills aiding adjustment to the new time zone is equivocal; Reilly *et al.* (1997a) found no benefit when travelling westwards, while others have reported beneficial effects in air force personnel. Even if the drug does help a person get to sleep, it does not guarantee a prolonged sleep, nor has it always been satisfactorily tested for subsequent residual effects on mental or physical performance. Sleeping pills might be counterproductive if given at the incorrect time; a prolonged sleep at the time an individual feels drowsy – presumably when they would have been asleep in the time zone just departed from – should be avoided, since it anchors the rhythms to this time zone and so hinders adjustment to the new time zone (Minors and Waterhouse, 1981b).

After travel

General lifestyle. In general, the advice is to adjust the sleep–wake schedule and mealtimes as rapidly and fully as possible to those at the destination. This might include sight-seeing in the first day or so after arrival, but there are potential problems with this as regards light exposure, problems that will be considered later. For athletes, training sessions should not be all-out efforts in the first days after arrival; skills requiring fine coordination are likely to be impaired and this might lead to a loss of confidence and even to accidents or injuries if, for example, games players conducted sessions with a ball too strenuously. Having at least one friendly match or competitive event during the initial period after arrival – that is, before the end of the first week – has been advocated (Reilly *et al.*, 1997b).

Subject to these caveats, exercise for athletes is recommended since it helps them to prepare psychologically for competition. In the days immediately after the flight, there will be a window of time during the day when the period of waking in the time zone just left overlaps the period of waking in the new time zone. This window should be used for training practices in the first few days, although it is prudent that training intensity should be lighter than normal.

General lifestyle and exposure to light can be used to promote adjustment of the body clock by a delay or an advance. For example, when the arrival time of a group of athletes travelling eastward from the UK to Oceania (Australia and New Zealand) was in the early morning, they were allowed to sleep on arrival until midday, and then allowed outdoors in the afternoon (Reilly and Mellor, 1988). When the arrival time was in the late afternoon, they were encouraged to be outdoors rather than to sleep. In both cases, this regimen of exposure to natural daylight in the later part of the day would promote a phase advance of the body clock. Conversely, athletes travelling westward to Oceania from the UK have followed a behavioural regimen that favoured a phase delay of the body clock.

Promoting sleep, alertness and physical performance. Drugs have been used to promote sleep in the new time zones, important among which are the benzodiazepines (Redfern, 1989; Donaldson and Kennaway, 1991; Atkinson *et al.*, 1997; Stone and Turner, 1997). However, Reilly *et al.* (1997a) found no beneficial effects of temazepam in Olympic athletes travelling from London to Florida (five time zones to the west). In such a case, it seems that disembarkation in the late evening local time following a westward flight (having had an extended period of wake time) might help one to get to sleep, thereby diminishing any potential effect of the drug.

In practice, the problem is also one of ensuring that travellers are fully awake and alert the next day. Most of the benzodiazepines are associated with residual effects on alertness and psychomotor performance, though temazepam may be less problematic in this respect since its half-life is shorter. There has been interest recently in zolpiden, which has a short half-life and affects short-term memory less than other drugs of this group (Jamieson *et al.*, 2001; Suhner *et al.*, 2001). In their review, Stone and Turner (1997) concluded that 'advice on the use of hypnotics requires further clarification', but they acknowledged that the general usefulness of such drugs was beyond doubt.

Promoting alertness has been investigated less, but the athlete is warned that many of the substances that have been used are banned compounds. Akerstedt and Ficca (1997) considered amphetamines, caffeine, the $\propto_1$-adrenoceptor agonist, Modafinil, and Pemoline, a drug with dopamine-like properties. These drugs sustain alertness and performance at mental tasks during extended periods without sleep, but they also reduce the ability to initiate and sustain sleep. Care must be taken in deciding when to take the drug, therefore. The use of amphetamines and caffeine can both lead to drug dependence; in addition, amphetamines can compromise decision making and caffeine can cause heart arrhythmias. By contrast, Pemoline and Modafinil have less deleterious effects upon complex mental tasks and have far less potential for drug abuse.

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There is a study (Lagarde *et al*., 2001) that indicates that caffeine might be of value to travellers; the decline in grip strength of the dominant hand that took place after an eastward flight across seven time zones was opposed by slow-release caffeine. The mechanism responsible for this is unclear, but another report, based on the same study, indicated that the caffeine had promoted a more rapid adjustment of the body clock to the new time zone (Pierard *et al*., 2001). Slow-release caffeine has been shown to decrease daytime sleepiness after an eastward flight across eight time zones, but also to affect some aspects of sleep quality (Beaumont *et al*., 2004)

Further work is required before any firm recommendation can be made regarding the use of drugs to promote alertness after time-zone transitions, and the warning regarding their use by athletes is repeated.

Clock-shifting: general principles. The main advice for travellers is to aim to promote adjustment of the body clock to the new time zone. Several methods have been suggested, differing in their practicality, the scientific evidence in support of them and in potential side-effects.

Promoting adjustment of the body clock can be by advance or delay, and each alternative requires a different timing of treatment. For time-zone transitions of up to 8 h to the east, a phase advance of the body clock should be promoted, and for time-zone transitions to the west, a phase delay. For journeys to the east through nine or more time zones, the position is more complex, since adjustment could be by advance or delay (Gundel and Wegmann, 1989). In practice, delaying the phase of the body clock can be more convenient for the individual concerned because, this way (see above), the time of minimum temperature and poorest performance delays into the evening rather than advances into the morning.

Redfern (1989) stressed the unlikelihood of developing a 'jet lag pill', since the circadian system possesses multiple input and output signals to and from the body clock. Dawson and Armstrong (1996) listed many classes of drugs that had the capacity to modify the phase of the circadian system ('chronobiotics'), but many of these have been tested only on animals.

Non-pharmacological methods have been developed also. They include nutritional, environmental and behavioural measures, and often amount to strengthening the *zeitgebers* in the new time zone. Several other methods have been reported, often anecdotally and in the popular press rather than with evidence based upon scientific tests. These methods include chiropractic (which was found to be ineffective in a study by Straub *et al*., 2001) and other methods – placing one's feet in paper bags, massage, herbal remedies and aromatherapy – that do not appear to have been rigorously tested.

Attention here will be directed mainly to the two methods that have been tested scientifically and shown to be effective. These are melatonin and bright light.

Use of melatonin. In normal circumstances, melatonin from the pineal gland is secreted into the bloodstream between about 21:00 and 07:00 h, and can be regarded as a 'dark pulse' or 'internal *zeitgeber*' for the body clock (Arendt, 1992). Melatonin capsules taken in the evening by local time in the new time zone (Lino *et al*., 1993; Arendt and Deacon, 1997) reduce the symptoms of jet lag, the effect having been confirmed in both sexes, after flights in either direction, and at whatever time the flight takes place. Recent reviews attesting to melatonin's efficacy have appeared (Haimov and Arendt, 1999; Herxheimer and Petrie, 2002).

Not all studies have shown that melatonin significantly ameliorates the problems associated with time-zone transitions (Spitzer *et al*., 1999; Edwards *et al*., 2000). These studies involved shifts eastwards across six or ten time zones (Fig. 5). Exactly why these studies should have produced negative results is unclear, but several possibilities exist. First, in the study of Edwards *et al*. (2000), the participants were sports scientists and elite sportspersons preparing for the Olympic Games. All were extremely highly motivated, and determined not to be adversely affected by any negative effects associated with travel. Such a degree of motivation might be an inappropriate background against which to measure the effect of a treatment for the debilitating effects of jet lag. Second, in both studies, as in most others, the participants' exposure to natural lighting was not controlled. There is evidence that melatonin administration cannot counteract phase shifts produced by light given at the same time (Krauchi *et al*., 1997). It is possible, therefore, that exposure to the light–dark cycle might have overwhelmed any effects due to melatonin. Third, there appears to be considerable inter-individual variability in the response to time-zone transitions, and this might account for the lack of statistical significance associated with the results in shown Fig. 5, even though the daily means of the amounts of jet lag indicated a beneficial effect of melatonin.

In some studies, melatonin ingestion has been found to exert beneficial effects in addition to improving sleep (for a review, see Atkinson *et al*., 2003). Thus, it has been measured: to accelerate adjustment of core temperature and urinary variables in a laboratory simulation; to decrease errors in a vigilance task in a field study upon aircrew; to improve mental performance in a simulation study; to accelerate adjustment of the melatonin rhythm in a field study; to accelerate adjustment of cortisol and melatonin rhythms in field study conditions; to maintain some aspects of physical

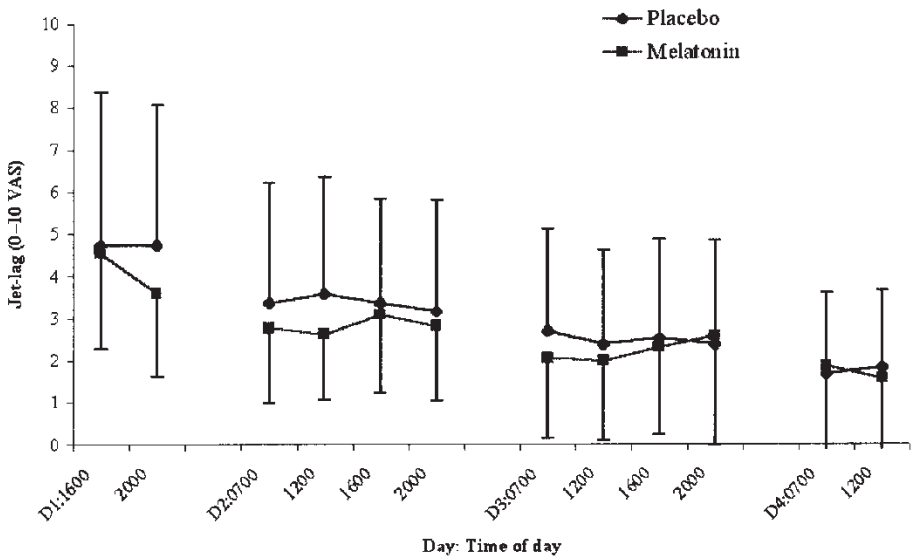


Fig. 5. Mean ($\pm$ *s*) jet lag (measured on a visual analogue scale, VAS) in a control group (placebo, $n = 17$) and an experimental group (melatonin, $n = 14$) during the first days after a flight from the UK to Australia (based on Edwards *et al.*, 2000).

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performance in the field; and to accelerate adjustment of blood pressure in a single male flying from Milan to Houston. In other studies, by contrast, melatonin has been found not to change core temperature or mental performance in the same way as it affected sleep and assessments of jet lag.

In addition to these reports of the value of melatonin to the traveller, the following points might be borne in mind (Waterhouse *et al.*, 1997; Reilly *et al.*, 1998):

A The mechanisms by which melatonin acts are uncertain: it might act as a hypnotic or as a chronobiotic. The hypnotic effect is probably related to the fall of core temperature that results from the vasodilatation caused by melatonin (Satoh and Mishima, 2001; Atkinson *et al.*, 2003). The possible chronobiotic action stems from the demonstration (Lewy *et al.*, 1998) that, in individuals living normally, phase advances are produced by administration between 13:00 and 01:00 h (peak advances at about 19:00 h) and delays by administration between 01:00 and 13:00 h (peak delays at about 07:00 h). To maximize this chronobiotic effect after a time-zone transition, melatonin would need to be taken at about 23:00 h after a westward flight across eight time zones, and at about 03:00 h after a flight across the same number of time zones but to the east. In practice, taking melatonin at the recommended time (about 2 h before sleep onset is required, see Haimov and Arendt, 1999) does not, therefore, appear always to be ideal for the chronobiotic effect.

B 'Jet lag', as defined in these studies, has concentrated on subjective symptoms and on the ability to sleep. It is unclear whether there are always comparable improvements in mental and physical performance, and this is an important issue for athletes. However, in one study performed on persons who were living normally and had not undergone a time-zone transition, melatonin capsules were ingested in the early evening (Atkinson *et al.*, 2001). The following morning, the participants did not report any reduction in alertness, mental performance or physical performance compared with the occasion when they took a placebo.

C Before melatonin can be recommended for extended use, in aircrew or athletes for example, it is necessary to establish that there are no adverse long-term effects. The lack of data pertaining to this issue has been stressed (Guardiola-Lemaitre, 1997); further work is urgently required (Herxheimer and Waterhouse, 2003).

There is increasing evidence for the use by travellers, including elite athletes, of melatonin, with little indication of serious side-effects. Individuals are recommended to take medical advice before taking melatonin for the first time, and not to do so if young, pregnant or lactating, or if an important event is to take place immediately after arrival in the new time zone (Reilly *et al.*, 1998). In addition, it has been stressed that the use of a 'standard dose' of melatonin is far less preferable than using one tailored to the individual (Manfredini *et al.*, 2000). Subject to these constraints, melatonin appears to be a valuable substance in combating jet lag.

Studies of melatonin analogues are beginning to appear in the scientific literature (see Nickelsen *et al.*, 2002, for example). While their short-term efficacy has

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been demonstrated, it will be some years before any side-effects due to chronic use can be assessed.

Clock shifting effects of bright light. The effectiveness of natural light in adjusting the body clock in humans has already been described and there are tables (see Table 4) and computer programs available to provide advice to travellers on when to attempt exposure to, or avoidance of, bright light. They are based on the observations that bright light in the morning (05:00–11:00 h) on body time advances the clock and bright light in the evening (22:00–04:00 h) on body time delays it (Czeisler *et al.*, 1989; Minors *et al.*, 1991). As a supplement to this treatment, there are also times when light should be avoided (those times which produce a shift of the body clock in the direction opposite to that desired).

On subsequent days, when partial adjustment of the body clock has occurred, the individual is then advised to alter the timings of light exposure and avoidance towards the local light–dark cycle by 1–2 h per day. In this way, the traveller's exposure to light gradually becomes synchronized with that of the locals.

The times of light-seeking and light-avoidance need not concur with the natural alternation between daytime and night. Avoiding bright light, by staying indoors or wearing dark glasses, is comparatively easy; seeking bright light is more of a problem. 'Light boxes' that mimic sunlight in spectral emission and brightness are available, but they are normally too cumbersome. What are now becoming available are visors, portable light sources with their own power (Boulos *et al.*, 2002). However, the demonstration that light of an intensity found domestically can alter the phase of the body clock (Boivin *et al.*, 1996; Waterhouse *et al.*, 1998; Boivin and James, 2002) might obviate the need for such light sources.

Adjusting as fully as possible to the lifestyle and habits in the new zone, and so being exposed to the new natural light–dark cycle, would seem intuitively to be the best remedy. This intuition is borne out by the example of a westward flight through eight time zones. To delay the clock requires exposure to bright light at 21:00–03:00 h *body time* and avoidance of it at 05:00–11:00 h *body time* on the first day. By new local time (see Table 4), this becomes equal to 13:00–19:00 h for bright light exposure and 21:00–03:00 h for bright light avoidance, staying indoors in dim light. (On the second day in the new time zone, light exposure at about 14:30–20:30 h and light avoidance at about 22:30–04:30 h are required, and so on.) Exposure to the new local day–night cycle – by getting 'out and about' and mixing with the local people – would provide this. For light after a westward shift, there is no need, therefore, to deviate from the timing of natural light.

In contrast, after a flight to the east through eight time zones, when a phase advance of the body clock is advised, this advice becomes inappropriate. The regimen required (see Table 4) is for bright light exposure at 05:00–11:00 h body time (13:00–19:00 h local time) and dim light (light avoidance) at 21:00–03:00 h body time (05:00–11:00 h local time). (On subsequent days, light exposure and avoidance can then be made earlier about 1–2 h per day until they coincide with natural lighting.) Importantly, morning light for the first day or so after the flight would be unhelpful in this case, as it tends to make the clock adjust in the wrong direction – though afternoon and evening light is fine. In this case, therefore, getting 'out and about' and mixing with local people needs to be delayed until the afternoon, the morning being spent indoors. This result is one explanation of why adjustment to phase advances is more difficult to achieve than is adjustment to phase delays.

A choice exists with regard to whether one uses light exposure and avoidance to delay or to advance the body clock when flights to the east cross more than nine time zones (Gundel and Wegmann, 1989). For example, after a flight to the east coast of Australia from the UK (an

Table 4. The use of bright light to adjust the body clock after time-zone transitions (based on Waterhouse *et al.*, 1997)

	Bad local times for exposure to bright light	Good local times for exposure to bright light
Time zones to the west		
4 h	01:00–07:00 [a]	17:00–23:00 [b]
8 h	21:00–03:00 [a]	13:00–19:00 [b]
12 h	17:00–23:00 [a]	09:00–15:00 [b]
16 h	13:00–19:00 [a]	05:00–11:00 [b]
Time zones to the east		
4 h	01:00–07:00 [b]	09:00–15:00 [a]
8 h	05:00–11:00 [b]	13:00–19:00 [a]
10–12 h	Treat these as 14–12 h to the west, respectively [c]	

[a] This will advance the body clock. [b] This will delay the body clock. [c] This is because the body clock adjusts to large delays more easily than to large advances.

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eastward flight through 10 time zones), Table 4 can be used either by treating the flight as an eastward one across 10 time zones or a westward one across 14 time zones. Where the choice is up to individuals, they must stick to one regimen throughout the process of adjustment.

When this choice between phase advancing or phase delaying the body clock exists, travellers would be advised to select the light exposure–light avoidance regimen that better fits in with their plans. Taking an eastward flight across 10 time zones as an example, on arrival the time of minimum body temperature, poorest performance and maximum fatigue will be at about 15:00 h local time. Arrival in daylight in the morning by local time (which will be before the temperature minimum) will promote a phase delay of the body clock, unless the light is avoided and dark sunglasses are worn until the shelter of a dimly lit hotel room is reached. By contrast, arrival in daylight in the late afternoon or early evening by local time (which will be after the temperature minimum) will promote a phase advance of the body clock, unless the bright light is avoided. If a phase advance of the body clock is being attempted, then the times of minimum body temperature, poorest performance and maximum fatigue will advance through the afternoon and morning on subsequent days. The effects of jet lag will be noticed for several days, therefore. If, by contrast, a phase delay of the body clock is attempted, then the times of poorest performance and so on will now delay through the afternoon and evening, and fit in better with relaxing at these times (Waterhouse *et al*., 2002a).

Clock shifting: diet and exercise. Other means to promote adjustment of the body clock have been proposed for humans. They include interventions such as feeding–fasting and physical activity–physical inactivity cycles.

The 'feeding hypothesis' (Leathwood, 1989) proposes that a high protein breakfast raises plasma tyrosine concentrations, and that this activates the arousal system of the body by promoting the synthesis and release of the neurotransmitters noradrenaline and dopamine. It is also proposed that a high carbohydrate evening meal raises plasma tryptophan (because the insulin release caused by such a meal does not clear tryptophan from the blood as readily as it clears tyrosine), and that this rise of plasma tryptophan promotes the synthesis and release of serotonin, a precursor of melatonin and a major neurotransmitter of the raphé nucleus (involved in sleep regulation).

Evidence favouring the feeding hypothesis is rather weak. It has been tested upon a group of military personnel undergoing an eastward time-zone transition of 9 h, but only small improvements in sleep and performance at mental tasks were observed, and the control and experimental groups appear to have been treated slightly differently (Graeber, 1982). Furthermore, experiments with rodents do not support the basic requirement of the hypothesis – that plasma concentrations of tyrosine or tryptophan are rate-limiting stages for neurotransmitter release (Leathwood, 1989).

A modification of this feeding hypothesis is the Argonne diet, which requires alternate feeding and fasting days before departure. A recent test of this diet upon military personnel (Reynolds and Montgomery, 2002) found evidence to support its efficacy, though how convenient it would be for the elite athlete, who often has special dietary requirements, is unclear.

In hamsters, the arousal of the central nervous system that is produced by physical activity (Redlin and Mrosovsky, 1997) promotes adjustment of the body clock to a shift of the light–dark cycle. Studies with humans have not yet provided convincing evidence for the phase-shifting effects of exercise (see Edwards *et al*., 2002), though whether this is because such a mechanism is inoperative, the amount of activity needs to be greater or participants must be more 'excited' by exercise than is normally the case is unclear. Nevertheless, to combine appropriately timed exposure to bright light with activity outdoors, and bright light avoidance with staying indoors and relaxing in dimmer light, would appear worth considering (Shiota *et al*., 1996). It has recently been reported (Cardinali *et al*., 2002) that a combination – of ingestion of melatonin at bedtime, of exercise and bright light exposure at 08:00–11:00 h and 13:00–16:00 h, and of avoidance of exercise and bright light at other times – enabled elite football players to adjust their sleep–wake cycle more rapidly and not to experience sleep difficulties following a westward flight across 12 time zones. However, no control group seems to have been used and there was no measurement of physical performance.

A checklist of dealing with jet lag is given in Table 5. Estimates of how long jet lag will last, if the ameliorative measures are practised, are given in Table 6.

Overview

Elite athletes and games players must be prepared to travel long distances to take part in competitions. Any long-haul flight involves planning, a change to an individual's training schedule and the need to sit in cramped conditions for extended periods of time. Travel fatigue – a transient sense of being 'travel-weary' – results from long-haul flights, but the effects can be minimized by suitable preparations for the journey and by taking appropriate precautions during the flight.

If the journey also involves crossing several time zones rapidly, then 'jet lag' is experienced for several days after the flight. This malaise manifests itself as a

Table 5. Checklist for dealing with jet lag (based on Waterhouse *et al.*, 2002b)

1. Check if the journey is across sufficient time zones for jet lag to be a problem. If it is not, then it is necessary only to refer to advice on overcoming travel fatigue (Table 1)
2. If jet lag is likely, then consider if the stay is too short for adjustment of the body clock to take place (a stay of less than 3 days). If it is too short, then remain on home time, and attempt to arrange sleep and activities to coincide with this as much as possible
3. If the stay is not too short (3 days or more) and it is wished to promote adjustment, then consider ways of reducing jet lag. Advice relates to:
   - Before the flight
   - During the flight
   - After the flight

   The most important advice relates to after the flight
4. Advice for promoting adjustment concentrates on:
   - Sleep and melatonin
   - Exposure to, and avoidance of, bright light (see Table 4)
   - Behavioural factors

Table 6. Estimates of how long jet lag will last (based on Waterhouse *et al.*, 2002b)

Westward flights		Eastward flights	
Time zones crossed	Days to adjust	Time zones crossed	Days to adjust
0–3	0 [a]	0–2	0 [a]
4–6	1–3	3–5	1–5
7–9	2–5	6–8	3–7
10–12	3–6	9–11	4–9

[a] Day of rest recommended to recover from travel fatigue.

group of symptoms; individuals feel fatigued and 'below par', and suffer negative effects with regard to mood and the abilities to train and perform at peak levels. Many of these effects can be ascribed to the slow adjustment of the body clock to the new time zone, as a result of which the circadian rhythms of sleep propensity and mental performance are no longer in synchrony with the new local time. It is important that the athletes and their coaches realize the debilitating effects of jet lag, and adjust their training schedules and expectations accordingly.

Current understanding of the properties of the body clock enables advice to be given to minimize the effects of jet lag. The measures centre on promoting sleep and adjustment of the body clock to the new time zone. Melatonin ingestion and bright light exposure–avoidance are most important regimens to fulfil these aims.

The management of sleep is important also for those who undertake tasks involving extreme endurance, such as transatlantic rowers. It is important also in extreme marathon sports that continue for days or weeks.

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Commentary

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As air travel becomes increasingly important in the world of competitive sport, both travel fatigue and cabin health concerns pose new challenges for athletes and their advisers alike.

To ensure optimum performance, travel fatigue must be actively managed, but an effective education programme needs to provide readers with sufficient understanding of the underlying physiology to appreciate fully the impact of travel on sleep, performance and general well-being. Waterhouse *et al.* attempt to address this need and I have comments regarding some of the advice proffered.

The term 'travel fatigue' as a generic concept comprises two main components, the stress effects that stem from the physical and psychological aspects of the flight itself (Carruthers *et al.*, 1976) and those effects that are a product of the need to reset the biological clock (Winget *et al.*, 1984). Travel fatigue and jet lag can therefore be considered together and not as separate entities.

The issue of dehydration and air travel remains a popular concern. The cabin environment does have less humidity than we normally experience in Northern Europe. However, it is no different to the environ-

mental conditions in many parts of the world. Travellers, provided they have normally functioning renal systems, will compensate for the environment and, although their skin and lips may feel superficially dry, they will not become dehydrated (Macnamara and Nicholson, 1969). Management on board should therefore be symptomatic and passengers should consume non-caffeinated, non-alcoholic beverages for comfort.

The authors mention the important issue of deep vein thrombosis in susceptible individuals, and they refer to the condition as 'economy class syndrome'. This term was dismissed as inappropriate by the House of Lords Select Committee on Science and Technology report on Air Travel and Health in 2000 (House of Lords Select Committee on Science and Health, 2000). The recommendation was to use the more accurate term 'traveller's thrombosis'.

The advice of the authors is well presented and clear, although there are a number of generalizations. An example is the recommendation that travellers should set their watches to destination time on boarding the aircraft and avoid in-flight sleep when it is not nighttime at the destination. The former may have no impact on some individuals' ability to manage their travel fatigue. Indeed, many frequent travellers, myself included, keep their watch on home time when travelling and use it as a key reference point in the first few days overseas when the circadian rhythms are adjusting to local time. The issue of when to sleep is complex, as sleep itself is so important in the effective management of circadian desynchronization. During a daylight flight to the USA, for instance, a 2-h nap during the natural circadian low in the afternoon will enable the passenger to cope better with the lengthened day, particularly if travelling to the west coast, where the late afternoon arrival time will be the early hours of the following morning to the traveller.

Despite these comments, I believe this review is a useful summary of important information in relation to the effects of travel fatigue and is worthwhile reading for both athletes and advisers.

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